Problem

Problem · Square roots of a complex number. We look for the complex numbers z such that $z^2 = 3 + 4i$. Let $z = x + iy$, with x and y real.

1. Prove that $z^2 = 3 + 4i$ is equivalent to the system $x^2 - y^2 = 3$ and $2xy = 4$.
2. Prove that if $z^2 = 3 + 4i$, then $\left(x^2 + y^2\right)^2 = \left(x^2 - y^2\right)^2 + (2xy)^2$, and deduce that $x^2 + y^2 = 5$.
3. Deduce the two solutions of the equation, then check them.
4. By the same method, determine the complex numbers whose square is $-5 + 12i$.
5. Consider the equation $z^2 - 3z + 3 - i = 0$.
   (a) Prove that it can be written $\left(z - \frac{3}{2}\right)^2 = \frac{-3+4i}{4}$.
   (b) Determine the complex numbers w such that $w^2 = -3 + 4i$.
   (c) Deduce the solutions of the equation and check their sum and their product.